Mohammed Zuhaib Damudi

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Education

Northeastern University Sep 2025 – Aug 2027
Master of Science in Computer Science *Boston, MA*
- **Coursework:** Algorithms, Database Management Systems, Foundations of Artificial Intelligence, Cloud Computing

Manipal Institute of Technology Jul 2019 - Jul 2023
Bachelor of Technology in Computer Science and Engineering, Minor Specialization in Big Data Analytics *Manipal, India*
- **Research Thesis:** "Single-Step Sampling Approach for Unsupervised Anomaly Detection of Brain MRI Using Denoising Diffusion Models" - *International Journal of Biomedical Imaging, Vol 2024, Issue 1*; DOI: 10.1155/ijbi/2352602

Experience

MassDOT Jul 2026 – Dec 2026
HR Business Intelligence Co-op *Boston, MA*
- Analyze multi-year survey and employee data for an organization of **4,000+ employees** to identify workforce trends and help leadership make informed staffing decisions.
- Use statistical analysis and machine learning models to test **100+** factors and identify key drivers of engagement.
- Build a Power BI reimbursement dashboard with custom DAX measures that flags mismatched reimbursement amounts across ServiceNow and Snowflake before payroll processing.
- Fulfill stakeholder data requests by combining timesheet and other HR data from multiple sources into custom reports.

Shell Aug 2023 – Aug 2025
Software QA Engineer *Bengaluru, India*
- Ran performance tests across **10+ projects** in Finance, Workforce Management, and IAM, simulating up to **200 requests per second** and **3,000 concurrent users** to find bottlenecks before production.
- Replicated real-world production workloads with live data loads across SAP systems (CFIN, BW, PaPM), exposing memory and CPU bottlenecks that caused system crashes under load.
- Identified redundant API calls on a web application where endpoints were being triggered multiple times per action, cutting unnecessary network requests by over **50%**.
- Contributed to a patent-filed *AI-powered code converter tool (GPT-4o)* that automated ~**80%** of manual programming effort.

Questt AI May 2022 – Jul 2022
Data Engineer Intern *Bengaluru, India*
- Designed and implemented an ETL pipeline in Apache Airflow that extracted and flattened over **1 million** heavily nested MongoDB documents into a structured relational schema in Google BigQuery.
- Built the company's primary data warehouse in BigQuery, replacing a third-party vendor and giving analysts direct SQL access for ad-hoc querying and reporting.

Projects

Publication - Anomaly Detection in Brain MRI using Generative AI | *Python, PyTorch, Matplotlib* Jan 2023 - Jul 2023
- Proposed a novel single-step sampling approach for brain tumor detection using DDPMs, reducing inference time from ~**50s** to ~**0.24s** per image. Published as first author in Wiley Online Library.
- Introduced a partial diffusion technique to generate healthy brain MRI, detecting anomalous regions with **80%** accuracy.
- Achieved a **66% improvement** in segmentation metrics (IoU, Precision, Recall, AUC) over the iterative sampling approach.

Runnalytics - Strava Running Analytics Dashboard | *Python, NumPy, Pandas, Scikit-learn, Flask* Dec 2025 - Jan 2026
- Processed Strava running exports containing heart rate, elevation, cadence, and GPS data, extracting **30+ performance metrics** and generating pace improvement trends and PR tracking.
- Developed an interactive dashboard with synchronized chart-to-map hover tracking, city-based route filtering via reverse geocoding, and dynamic elevation and heart rate visualizations.

Habitual - Mobile Habit Tracker Application | *Flutter, SQLite* Jan 2022 - Mar 2022
- Built and published a cross-platform habit-tracking app in Flutter on the Google Play Store, reaching **100+ downloads**.

Technical Skills

Languages: Python, C, C++, Java, JavaScript, HTML/CSS, Dart, SQL
Technologies: NumPy, Pandas, Matplotlib, PyTorch, Scikit-learn, OpenCV, MongoDB, MySQL, Power BI, Kibana, Git, Docker, Flutter, Google BigQuery, Apache Airflow, Apache JMeter, LoadRunner Enterprise, Azure Load Testing, Selenium, Postman
Concepts: Data Science, Data Engineering, Machine Learning, Big Data Analytics, Data Visualization, Anomaly Detection, Computer Vision, Generative AI, ETL Pipelines, Automation